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Semester: III	**Year : AY 2026-27**
Subject: Programming with Java	
ASSIGNMENT No : 5	
TITLE : Collection Classes and StringBuffer	

Problem Statement

Develop a Java application demonstrating the use of ArrayList, Vector, and StringBuffer for dynamic data storage, management, and string manipulation.

Learning Objectives

To implement dynamic data storage and string manipulation using ArrayList, Vector, and StringBuffer.

Programme

```
import java.util.ArrayList;
import java.util.Vector;

public class ToDoListDemo {
    public static void main(String[] args) {
        // ArrayList to store tasks
        ArrayList<String> tasks = new ArrayList<>();
        tasks.add("Complete Java assignment");
        tasks.add("Revise DSA in C");
        tasks.add("Go to the gym");

        // Vector - synchronized alternative to ArrayList
        Vector<String> completedTasks = new Vector<>();
        completedTasks.add("Attend Java lab");

        // remove a task from ArrayList
        tasks.remove("Go to the gym");

        // StringBuffer to build a display string efficiently
        StringBuffer taskDisplay = new StringBuffer();
        taskDisplay.append("Pending Tasks:\n");
        for (int i = 0; i < tasks.size(); i++) {
            taskDisplay.append((i + 1) + ". " + tasks.get(i) + "\n");
        }

        System.out.println(taskDisplay);
```

```
        System.out.println("Completed Tasks:");
        for (String task : completedTasks) {
            System.out.println("- " + task);
        }

        System.out.println("Total pending tasks: " + tasks.size());
    }
}
```

Output

```
C:\Users\Aryyan> java ToDoListDemo

Pending Tasks:
1. Complete Java assignment
2. Revise DSA in C

Completed Tasks:
- Attend Java lab
Total pending tasks: 2
```

Conclusion

Thus, a Java application was developed demonstrating the use of ArrayList, Vector, and StringBuffer for dynamic data storage and string manipulation. ArrayList was used to store and remove tasks dynamically, Vector was used to maintain a synchronized list of completed tasks, and StringBuffer was used to efficiently build a multi-line display string without creating multiple intermediate String objects. This helped in understanding the practical differences and use cases of these three classes.